

Q1

$$\begin{aligned}y &= 3 \sin \left[\frac{1}{2} \left(x - \frac{\pi}{6} \right) \right] - 4 \\ \frac{dy}{dx} &= 3 \cos \left[\frac{1}{2} \left(x - \frac{\pi}{6} \right) \right] \cdot \frac{1}{2} \cdot 1 \\ &= \frac{3}{2} \cos \left[\frac{1}{2} x - \frac{\pi}{12} \right]\end{aligned}$$

Given Ring $\mathbb{R}$